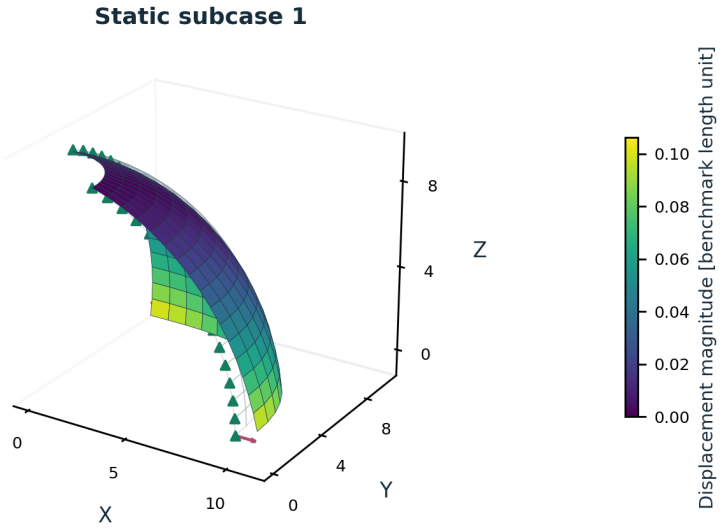
Pinched hemisphere with 18-degree opening

MH_hemispherical_shell | SOL 101 | Published shell benchmark

Quarter spherical shell; R=10, t=0.04, E=6.825e7, nu=0.3; polar angle 18 to 90 degrees.

Geometry / mesh	25 nodes; 16 elements. Quarter spherical shell; R=10, t=0.04, E=6.825e7, nu=0.3; polar angle 18 to 90 degrees.
Material / section	E=6.825e+07, nu=0.3; shell thickness=0.04. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Equator free; symmetry edges constrained; opposite radial point-force pair represented by +X and -Y unit forces at quarter equator.
Applied load	Quarter-model radial point forces: +1 in X and -1 in Y at the equator.



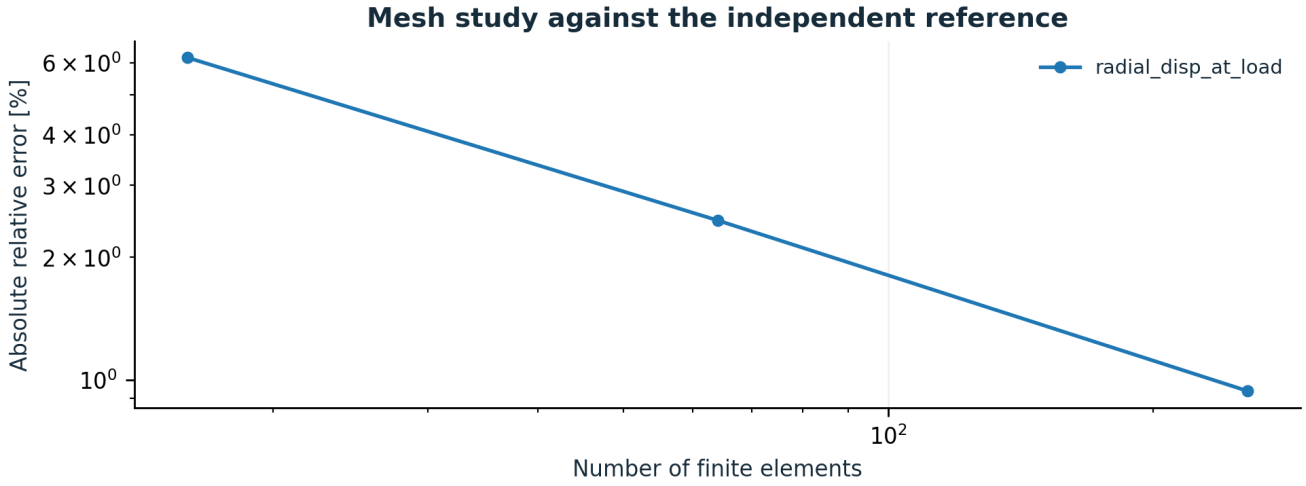
Shape amplified x11.3; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads. Finest companion shown.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
radial disp at load [benchmark length unit]	0.098748	0.093	6.1811%	PASS
N08 refined response [benchmark length unit]	0.09529	0.093	2.462%	PASS
N16 refined response [benchmark length unit]	0.093872	0.093	0.93755%	PASS

Evidence, convergence and interpretation



The figure shows the finest qualified companion; the table retains the supplied benchmark and available refinements. Each field dataset and binary file has its own input hash and run receipt.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Ko, Y.; Lee, Y.; Lee, P.-S.; Bathe, K.-J. (2017). Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Computers & Structures 193, 187-206. https://doi.org/10.1016/j.compstruc.2017.08.003

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv